

ANANTHA KRISHNAN K

AI / ML Engineer

ananthan0377@gmail.com | +91 90373 22474 | linkedin.com/in/anantha-krishnan-k | github.com/anantha037

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building production-grade multi-agent AI systems, RAG pipelines, and fine-tuned language models. Proficient across the full ML stack — from classical machine learning and deep learning to LLM engineering, MLOps, and Big Data. Demonstrated ability to architect and ship end-to-end systems, including a multi-agent Indian legal AI platform (LexShield AI) and AlignForge — a complete SFT-to-DPO alignment pipeline for Phi-3-mini with LLM-as-judge evaluation and models shipped to HuggingFace Hub. Seeking AI/ML Engineer roles where strong fundamentals meet production thinking.

TECHNICAL SKILLS

Languages	Python, SQL, Advanced SQL, Bash
ML / DL	Regression, Classification, Clustering, Ensemble Methods, Dimensionality Reduction, Time Series Analysis, Deep Learning, CNN, RNN, GANs, Transformers, NLP, Advanced NLP
Frameworks	TensorFlow, PyTorch, scikit-learn, OpenCV, spaCy, HuggingFace Transformers, PEFT, TRL, BitsAndBytes, pandas, NumPy, Matplotlib, Seaborn
LLM / AI Eng.	LangGraph, RAG Pipelines, ChromaDB, Groq, Gemini API, Multi-Agent Orchestration, Vector Databases, Prompt Engineering, RAGAS, QLoRA, LoRA
MLOps	MLflow, Weights & Biases, FastAPI, Docker, CI/CD for ML, Model Versioning, Logging & Monitoring, Git, Jupyter
Big Data	Apache Spark, Spark ETL & ML, Hadoop, PySpark
Cloud	Google Cloud Run, AWS EC2, AWS SageMaker, Heroku
Web / Other	Django, Advanced Django, Streamlit, REST APIs, DSA, Linux

EXPERIENCE

Machine Learning Engineer — Bootcamp <i>Brototype Kerala, India</i>	2025 – Present
- Intensive industry-focused ML Engineering bootcamp covering mathematical foundations through production-grade MLOps, Big Data, and LLM Engineering. - Built 8+ production-grade ML/AI projects including a multi-agent legal AI platform and a complete LLM alignment pipeline shipped to HuggingFace Hub.. - Capstone project: LexShield AI — production-grade multi-agent legal AI platform (see Projects).	
Data Science Intern <i>YBI Foundation Delhi</i>	Jun 2024 – Jul 2024
- Built a Movie Recommendation System using collaborative filtering to personalise user preferences. - Developed a Big Sales Prediction model using regression to forecast retail sales and optimise inventory strategy.	

PROJECTS

LexShield AI — Multi-Agent Legal AI Platform <i>LangGraph, RAG, ChromaDB, FastAPI, Gemini 2.0 Flash, Groq LLaMA 3.3 70B, InLegalBERT, spaCy, NetworkX, Python</i>	2025
- Architected a full-stack multi-agent legal AI system targeting Indian legal inaccessibility (50M+ pending court cases), ingesting 60+ Indian Acts across 22,185 semantic chunks. - Built LangGraph StateGraph with 6 agent nodes (retrieval, drafting, NER, translation, risk scoring, knowledge graph), intent classifier supporting 6 intents, and 5-turn session memory. - Implemented BM25 hybrid search with RRF fusion, LLM query rewriting, NVIDIA NIM reranking, and multi-document synthesis with citation grounding; achieved 77.4% GOOD evaluation score via RAGAS.	

- Fine-tuned InLegalBERT on synthetic legal data for document classification (8 categories); built clause-level risk scoring engine with 13 rules.
- Developed StatefulDraftingAgent generating 5 legal document types and TranslationAgent supporting 10 Indian languages via Groq.
- Built LegalKnowledgeGraph via NetworkX with IPC↔BNS section pair relationships auto-populated from corpus.

AlignForge — SFT + DPO Alignment Pipeline for Phi-3-mini

2025

QLoRA, PEFT, TRL, BitsAndBytes, DPO, HuggingFace Transformers, Weights & Biases, Groq, Kaggle T4 GPU

- Built an end-to-end instruction tuning and alignment pipeline for Microsoft Phi-3-mini (3.8B) — from raw pretrained model to a human-preference-aligned assistant.
- Implemented QLoRA (4-bit quantisation + LoRA adapters via PEFT) reducing trainable parameters from 3.8B to 20M, enabling full fine-tuning on a free-tier GPU without quality loss.
- Designed custom label masking during SFT on Dolly-15k — computing loss only on response tokens so the model learns to generate answers, not memorise instructions.
- Applied DPO alignment on Anthropic hh-rlhf (160K human preference pairs) with a frozen SFT reference model; tuned beta to control divergence from rejected responses.
- Evaluated with LLM-as-judge (LLaMA 3.3 70B via Groq) scoring helpfulness and safety — current industry standard over BLEU/ROUGE for open-ended generation.
- Tracked all experiments via Weights & Biases; shipped both SFT and DPO model checkpoints to HuggingFace Hub with full model cards.

Additional Projects: Multi-label text classification (DistilBERT + MLflow + Docker), LoRA image classifier (ViT + Grad-CAM), AIOps log anomaly detection (Isolation Forest + Streamlit), large-scale PySpark ETL & ML pipeline (9.3M NYC taxi trips, Hadoop + Hive). All available on GitHub: github.com/anantha037

EDUCATION

B.Sc (Hons) Computer Science

2021 – 2024

Hansraj College, University of Delhi Delhi

- CGPA: 7.838 / 10
- Member, ORDINATEUR — Computer Science Society; managed operations and sponsorships for college events.

CERTIFICATIONS & COURSES

- **Cognizant AI Job Simulation** (Forage) — EDA, ML model development, results communication.
- **BCG GenAI Job Simulation** (Forage) — AI-powered financial chatbot using 10-K/10-Q data with rule-based logic.
- **British Airways Data Science Simulation** (Forage) — customer review scraping, predictive modelling for buying behaviour.
- **The Data Science Course: Complete Bootcamp** — Udemy (Python, Statistics, ML, Deep Learning).
- **TCS iON Career Edge** — Young Professional (communication, business, and IT fundamentals).
- **High Performance Computing Workshop** — IIT Kharagpur (parallel processing, optimisation strategies).